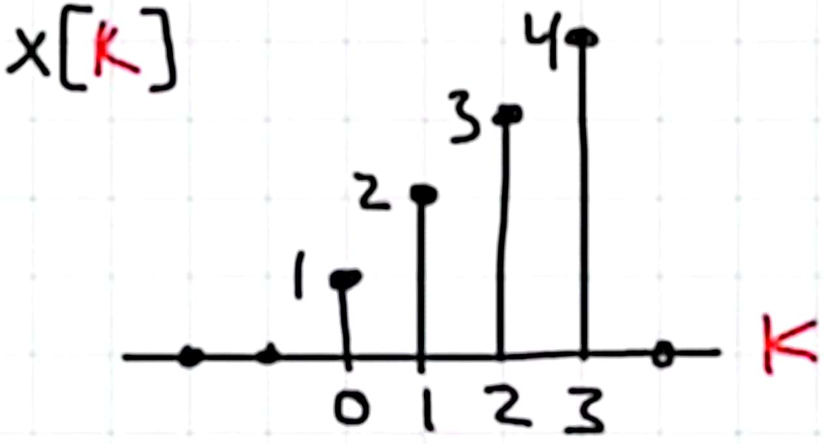
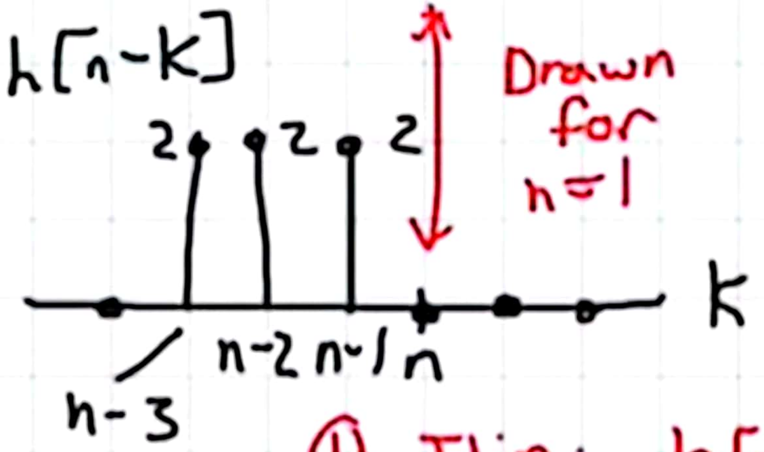
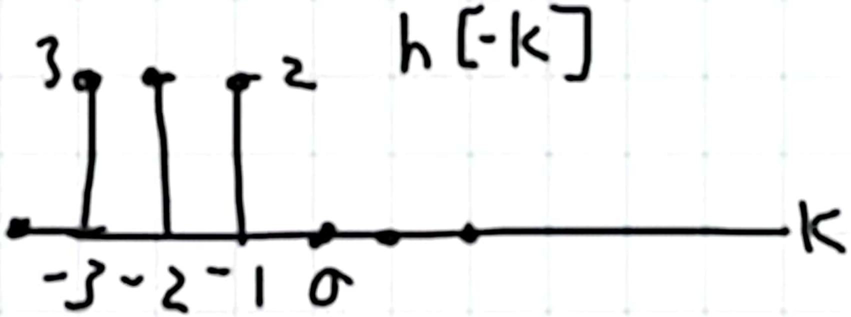
Find: $y[n] = h[n] * x[n] = \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]$







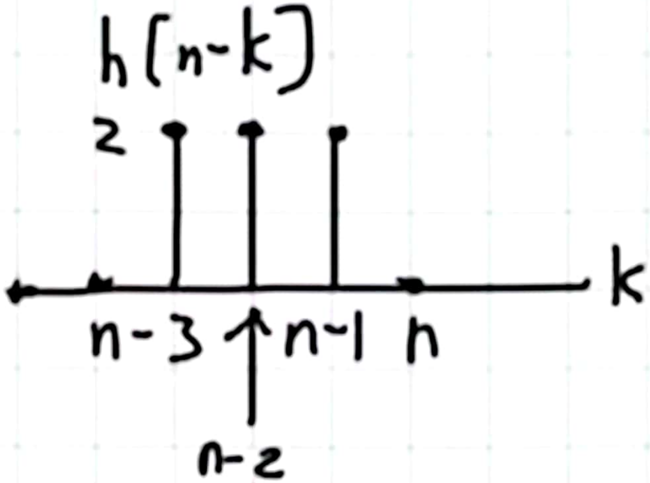
① Flip: $h[-k]$

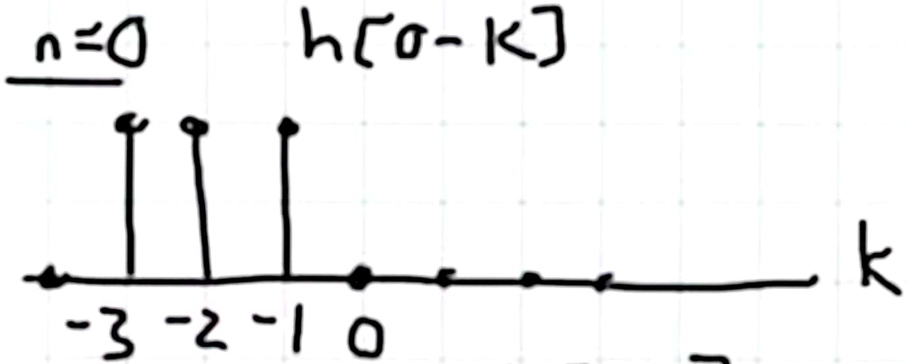
② Shift: $h[n-k]$

③ Multiply: $x[k]\,h[n-k]$

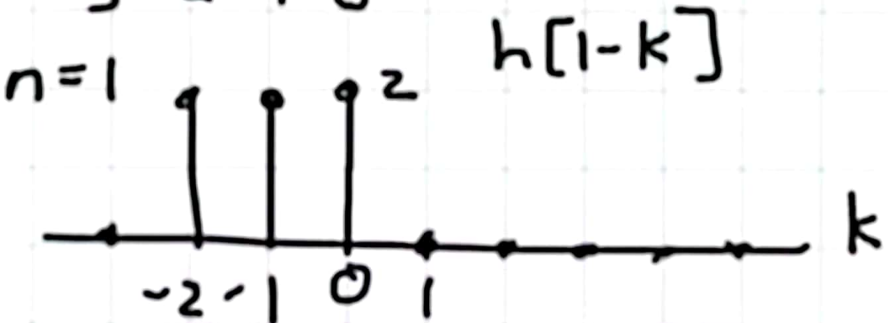
④ Add: $\sum x[k]\,h[n-k]$

Repeat 2-4 for all n

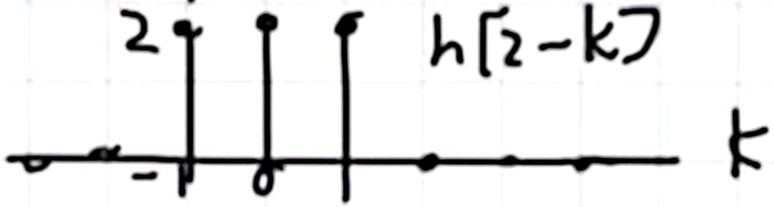




$$y[0] = \sum_k x[k]h[0-k]$$
$$= \sum_k 0 = 0$$

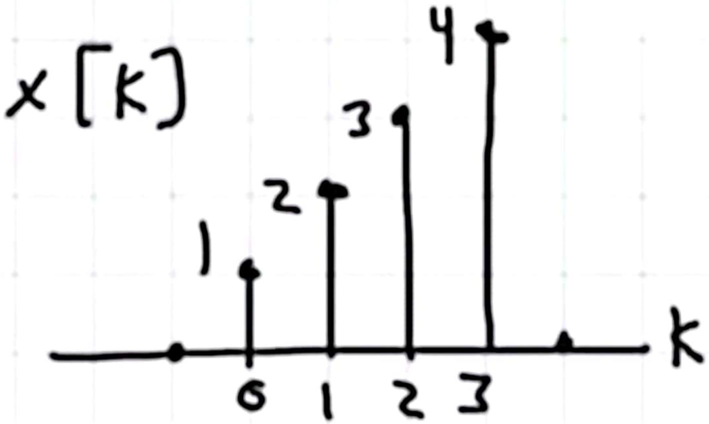


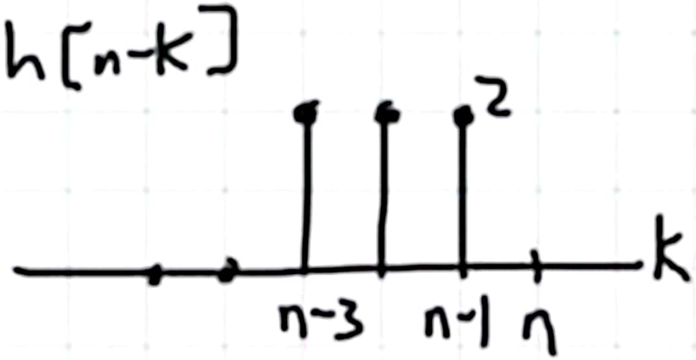
$$y[1] = \sum_k x[k]h[1-k]$$
$$= 1\cdot 2 = 2$$



$$y[2] = \sum_k x[k]\,h[2-k]$$
$$= 1\cdot 2 + 2\cdot 2 = 6$$

$$y[n] = \sum_{k} x[k]\,h[n-k]$$





Result →